

01 — Departure from CSR: $\tilde{\delta}$ and null tables

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1 Theory

Setting

- Window $W = [0, 1]^2$, fixed
- $\bar{n}$: expected number of points (deterministic model parameter)
- Pattern $x = \{x_1, \dots, x_n\} \subset W$; $n = |x|$ realised number of points (random around $\bar{n}$)
 - Poisson, Matérn II: $n/\bar{n} \in [0.90, 1.11]$ (5–95%)
 - Thomas, nested, LGCP: $n/\bar{n} \in [0.77, 1.25]$ (overdispersed, clusters arrive whole)

Estimator

- Ripley's K , isotropic edge correction, λ^2 estimated by $n(n-1)$:

$$\hat{K}_x(r) = \frac{1}{n(n-1)} \sum_{i \neq j} w(x_i, d_{ij}) \mathbf{1}\{d_{ij} \leq r\}, \quad w(x, d) = \frac{2\pi}{\text{length of } \partial b(x, d) \cap W}$$

- Centred L -function of a pattern x , and its model counterpart:

$$\hat{\phi}_x(r) = \hat{L}_x(r) - r = \sqrt{\hat{K}_x(r)/\pi} - r, \quad \phi(r) = L(r) - r \quad (= 0 \text{ under CSR})$$

- Evaluated on the radius grid: $\hat{\phi}_x = (\hat{\phi}_x(r_1), \dots, \hat{\phi}_x(r_{512})) \in \mathbb{R}^{512}$
- Radii $r_k = 0.25 k/512$, $k = 1, \dots, 512$ ($r = 0$ dropped: identically zero)

Radius set

Everything in this subsection is defined for fixed n . The test and $\tilde{\delta}$ only use radii in $R(n)$.

- At small r , $\hat{\phi}_x(r)$ is a function of a small count: the pairs of x closer than r . Under $\mathbb{P}_{0,n}$ we expect

$$\mu(r; n) = \frac{1}{2}n(n-1)\pi r^2$$

such pairs (edge-corrected)

- The maximum in M_n (next subsection) treats all radii alike only if each standardised deviation $(\hat{\phi}_X(r) - m_0)/s_0$ has roughly the same null distribution, i.e. $\approx N(0, 1)$
- Few expected pairs $\Rightarrow \hat{\phi}$ takes a handful of values; standardised deviation discrete and skewed, not $N(0, 1)$. Checked on the null curves at $n = 99$:
 - $\mu < 0.3$: heavy right tail, $\mathbb{P}(|z| > 3)$ 10 $\times$ Gaussian
 - $\mu \in [3, 10]$: still 3 $\times$ Gaussian
 - $\mu \geq 10$: Gaussian tail rate, skewness and kurtosis
- Size does not need them to be Gaussian (c_{95} is simulated). Two things do bind:

- Heavy-tailed radii win the null maximum too often $\Rightarrow c_{95}$ up (2.75–3.21 at $p = 10$, 2.87–3.28 at $p = 5$) $\Rightarrow$ 1–3 points of power lost at every other radius
- $c_{95}(n)$ must stay smooth in n , or the spline across n (Smoothing) cannot be fitted honestly
- The smoother is what fails first. Below $p \approx 5$ the raw c_{95} develops structure at large n that no spline follows (residual s.d. vs Monte Carlo noise; z against the fit):

p	c_{95} mean z^2 (expect ≈ 1)	residual s.d.	Monte Carlo noise
10	0.77	0.019	0.022
5	0.97	0.023	0.022
4	1.62	0.036	0.029
3	5.22	0.040	0.026

At $p = 3$ the excess is concentrated at large n (residual s.d. 0.056 for $n \geq 1000$) and locally correlated (acf 0.30): the max statistic's null distribution is lumpy there, so its 95% quantile moves in jumps. Covariates for the grid geometry do not absorb it (mean z^2 5.22 $\rightarrow$ 4.67 at best)

- Keep radii with at least p expected pairs, $p = 5$:

$$R(n) = \{r_k : \mu(r_k; n) \geq p\} = [r_{\min}(n), 0.25], \quad r_{\min}(n) = \sqrt{\frac{2p}{\pi n(n-1)}} \approx \frac{1.78}{n}$$

- p is chosen from null diagnostics alone (size, c_{95} smoothness), never from any alternative's power
- $r_{\min}$: 0.092 ($n = 20$), 0.018 ($n = 100$), 0.0022 ($n = 800$), 0.00089 ($n = 2000$). More points $\Rightarrow R(n)$ reaches down to finer scales

Null distribution

Everything in this subsection is defined for fixed n . In practice, we compute the null distribution at every n on a grid.

- Null $\mathbb{P}_{0,n}$: n i.i.d. uniform points on W (CSR conditioned on n); $\mathbb{E}_{0,n}$, $\text{sd}_{0,n}$ under it
- Generate 12 000 CSR clouds from $\mathbb{P}_{0,n}$
- Estimate null moments for $r \in R(n)$ using half of the CSR clouds:

$$m_0(r; n) = \mathbb{E}_{0,n}[\hat{\phi}_X(r)], \quad s_0(r; n) = \text{sd}_{0,n}[\hat{\phi}_X(r)]$$

- Using remaining half of CSR clouds, compute maximum deviation of each cloud x :

$$M_n(x) = \max_{r \in R(n)} \frac{|\hat{\phi}_x(r) - m_0(r; n)|}{s_0(r; n)}$$

- Determine critical value: $c_{95}(n) = 95\%$ quantile of M_n over these clouds (estimates the 95% quantile of $M_n(X)$, $X \sim \mathbb{P}_{0,n}$)
- Repeated on a grid of n ; m_0 , s_0 , c_{95} then smoothed across n (cubic splines in $\log n$) $\Rightarrow$ less Monte Carlo noise, defined at every n

Studentised global envelope (Myllymäki et al. 2017)

$$S(x) = \frac{M_n(x)}{c_{95}(n)}, \quad n = |x|$$

- Under the null, $\mathbb{P}_{0,n}(S > 1) = 0.05$ by definition of c_{95} (up to Monte Carlo error)
- Reject CSR at 5% iff $S > 1$

Distance $\tilde{\delta}$

$$\tilde{\delta}(\theta, \bar{n}) = \frac{1}{c_{95}(\bar{n})} \max_{r \in R(\bar{n})} \frac{|L_\theta(r) - r|}{s_0(r; \bar{n})}$$

- Same functional as S , applied to the closed-form L_θ at $\bar{n}$, uncentred ($m_0 = 0$)
- $\tilde{\delta}(\text{CSR}) = 0$ exactly
- $\tilde{\delta} = 1$: true curve on the rejection boundary $\Rightarrow$ power well above 5%; measured 0.64–0.78 (02), not 50%, because pattern noise adds to the maximum
- Units: null s.d.s, relative to the critical value
- Property of the model, known before simulation $\Rightarrow$ usable as a sweep coordinate
- At fixed shape, $\tilde{\delta}$ grows with $\bar{n}$ (s_0 shrinks)
- Depends on the cutoff when the departure lives below $r_{\min}$: a hard core of radius $R < r_{\min}$ enters $\tilde{\delta}$ only through the pair deficit it leaves above $r_{\min}$, diluted by the CSR count there, so its $\tilde{\delta}$ scales with p (Matérn II; 02)

2 Implemented pipeline

Code

src/cloudforger/classical/lfunction.py	$\hat{K}, \hat{L} - r$, weights, $L - r$ from $K - \pi r^2$
src/cloudforger/departure/config.py	config, n grid, paths
src/cloudforger/departure/simulate.py	null patterns $\rightarrow$ stored curves
src/cloudforger/departure/fit.py	curves $\rightarrow$ spline tables
src/cloudforger/departure/tables.py	tables $\rightarrow S, \tilde{\delta}$
src/cloudforger/departure/validate.py	size on fresh patterns
scripts/departure.py	CLI: <code>simulate</code> , <code>fit</code> , <code>validate</code>
configs/departure/	<code>config.yaml</code> , <code>tables.npz</code> , <code>report.json</code>

Configuration

- n grid: $\text{round}(20 \cdot 1.03^k)$ up to 2000 $\Rightarrow$ 153 values
- Per grid n : 6000 fit + 6000 calibration patterns
- Validation: 50 000 patterns, n log-uniform integer on $[20, 2000]$
- $p = 5$ pairs; $\alpha = 0.05$; knots: 8 interior (moments), 4 interior (c_{95})

Steps

1. **Simulate**: null curves for every grid n , stored
2. **Fit**: moments and c_{95} per n , splines in $\log n \rightarrow \text{tables.npz}$
3. **Validate**: size on fresh patterns at random n
4. **Relabel** (`scripts/relabel.py`): after a refit, recompute $\tilde{\delta}$ for every swept θ under the new tables; manifests keep `delta` (the sampling target) and `delta_tilde` (current, what regimes use)

Results

Check	Result	Reference
Fitted m_0 vs raw, mean z^2	0.94	≈ 0.92 (1 – params/grid)
Fitted s_0 vs raw, mean z^2	0.96	≈ 0.92
Max relative error of s_0	4.0%	—
Fitted c_{95} vs raw, mean z^2	0.97	≈ 0.95
c_{95} range; max s.e.	2.87–3.28; 0.030	—
Size, 50 000 patterns	0.0518 ($z = 1.8$)	0.05
Size by n band	0.049–0.056, all $ z \leq 2.4$	0.05
Size, 20 000 Poisson sweep patterns	0.0496	0.05
Centring gap max $ m_0 /s_0/c_{95}$	0.026–0.044	—

Estimator checks

- `tests/test_lfunction.py`:
 - Weights vs brute-force circle fraction: $|\Delta| < 10^{-3}$
 - Two points at distance 0.1: $\hat{K} = 0$ below, $= 1$ above
 - $\mathbb{E}\hat{K} = \pi r^2$ under binomial (400 patterns, within 3%)
 - One-off: agrees with the previous implementation (`baselines/vihrs.py`) to 10^{-9}
- Further statistical notes: `01_departure_notes.md`.